ELECTRICITY CONSUMPTION AND SENSE OF 'ENVIRONMENT' IN EVERYDAY PRACTICE: A CROSS SECTIONAL STUDY IN BANGLADESH

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INTRODUCTION

- Bangladesh is a developing country with extensive population growth and density. About 1265 people live per square kilometer in Bangladesh and it is home to around 160 million people.
- Because of this high population density electricity consumption is rising day by day. From 2000-2014 electricity consumption increased by 206%, and from 2000-2021 it reached to 526% (Debnath and Mourshed, 2022).
- The production of electricity in our country is mainly fossil fuel dominating; especially coal intensive. This leads the country towards huge GHG emissions and the severe effects of climate change. The projected per capita emissions of CO2 will rise up to 201 and 402 million tons by 2030 and 2050 with the UN projected population.
- At this point it has become significant to know about people's awareness and conception of electricity consumption concerning the environmental impacts.

OBJECTIVES

- To evaluate the awareness and perceptions of the respondents regarding electricity consumption.
- To demonstrate the conjugation among electricity consumption with various divisions and locations.
- To ascertain the respondents' daily use of efficient and expensive electronic appliances and their correlation with Geographical location.
- To appraise the knowledge of the respondents about climate change, sources of electricity generation, and causality of electricity generation for environmental degradation.

REVIEWING THE RELATED LITERATURE

ELECTRICITY CONSUMPTION AND ENVIRONMENTAL DEGRADATION

(Alam, Hassan and Sadekin, 2021)

- This article is about energy consumption, environmental degradation, and economic growth in Bangladesh covering the period from 1772-2018.
- A corresponding statistical Co-integration Test, optimal lag test, diagnostic test, and Granger causality test is done to have a suspected outcome.
- Statistics for electricity consumption per capita with per capita GDP and a long-run bi-directional nexus among economic growth, capital formation, electricity consumption, energy consumption, and carbon emission are found in the study.
- Although economic growth is not our major concern, we derived abundant knowledge about the effect of energy use and CO2 emission.

Alam, Hassan and Sadekin. (2021). Impact of Energy Consumption and Environmental Degradation on Economic Growth in Bangladesh. *Central Asia and the Caucasus*, 612-628

ELECTRICITY CONSUMPTION AND ENVIRONMENTAL DEGRADATION

(Debnath and Mourshed, 2022)

- This study highlighted the factors of how electricity generation in Bangladesh is leading Green House Gas (GHG) emissions and also delineated the future climatic intensity of Bangladesh.
- The power generation rate of 74 public and 85 privately owned power plant units is reviewed in the paper. Also highlighted is the distribution of different types of Powerplants in Bangladesh as Thermal power plants, Renewable Powerplants, and Hydro Powerplants in a map.
- Though it was a review paper and they don't do any research, we attained knowledge and concepts about how electricity generation is leading our country toward climatic vulnerability by emitting greenhouse gases and how we can have a reliable, sustainable, and reasonable electricity production without harming the environment.

Debnath and Mourshed. (2022). Why is Bangladesh's electricity generation heading towards a GHG emissions-intensive future? *Carbon Management*, 216-237.

ELECTRICITY CONSUMPTION AND ENVIRONMENTAL DEGRADATION

(Muntasir Murshed, et al., 2021)

- This paper is about Foreign direct investments for renewable electricity output and ecological footprints. To facilitate renewable energy transition and the environmental welfare of Bangladesh was the objective of the study.
- Unit root analysis, Cointegration analysis, and causality analysis methods is used to have the suspected outcome.
- From the regression and causality analysis, they found that Foreign Direct Investment (FDI) enhanced the REO (Renewable Energy Output) shares in Bangladesh. FDI makes a very important role in boosting a nation's ecological footprint, thus directly related to environmental degradation.
- The paper is not directly related to our objectives but we ontained a notion about the importance of renewable electricity sources to have a sustainable environment for the welfare of our country.

Muntasir Murshed, et al. (2021). Foreign Direct Investments, Renewable Electricity Output,. *Springer*, 33-78.

METHODOLOGY

PARTICIPANTS AND PROCEDURE OF DATA COLLECTION:

- The study started by collecting primary data through a structured Questionnaire Survey at the household level.
- There were a total of 300 respondents of age group 18-70 who successfully participated in the survey on electricity consumption and sense of 'environment'. The respondents are of different occupations and educational qualifications.
- This cross-sectional study includes respondents from all eight divisions in Bangladesh and includes the respondents' current location.
- Secondary data is collected from Bangladesh Power Development Board (BPDB).

MEASURES:

The aim of the study was to delineate the correlation between electricity consumption and people's 'sense' of the environment. The Structured Questionnaire Survey had several sections a)Socio-demographic b)Electric appliances and their use c)Conception of Electricity resources d)'Sense' of Environment e)Policy.

- **Socio-demographic measures:**

Information on socio-demographic variables including age(18-70), gender(Female, Male), family monthly income(<25,000BDT,25,000–50,000BDT,>50,000BDT), Occupation(Student, Government Job Businessman. Housewife. Private Job. Unemployed) Education (Postgraduate, Graduate, Below HSC, Below SSC, Illiterate, Primary Education, Others) Location of residence (City, Village, Upazila), Division(Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet).

- **Electric appliances and their use:**

Information on a)Expensive Appliances, b)Efficient Appliances, c)Monthly electricity bills, d)Electricity Availability, e)satisfaction, f)Reason for such satisfaction, g)Reasons behind the increased use of Electricity, h)Role of seasonal changes, i)Most electricity used to season, j)Concerned about the abuse of power, k)The reason to worry is collected to know about the consumption of electricity by the respondents.

MEASURES:

- **Conception on Electricity resources**

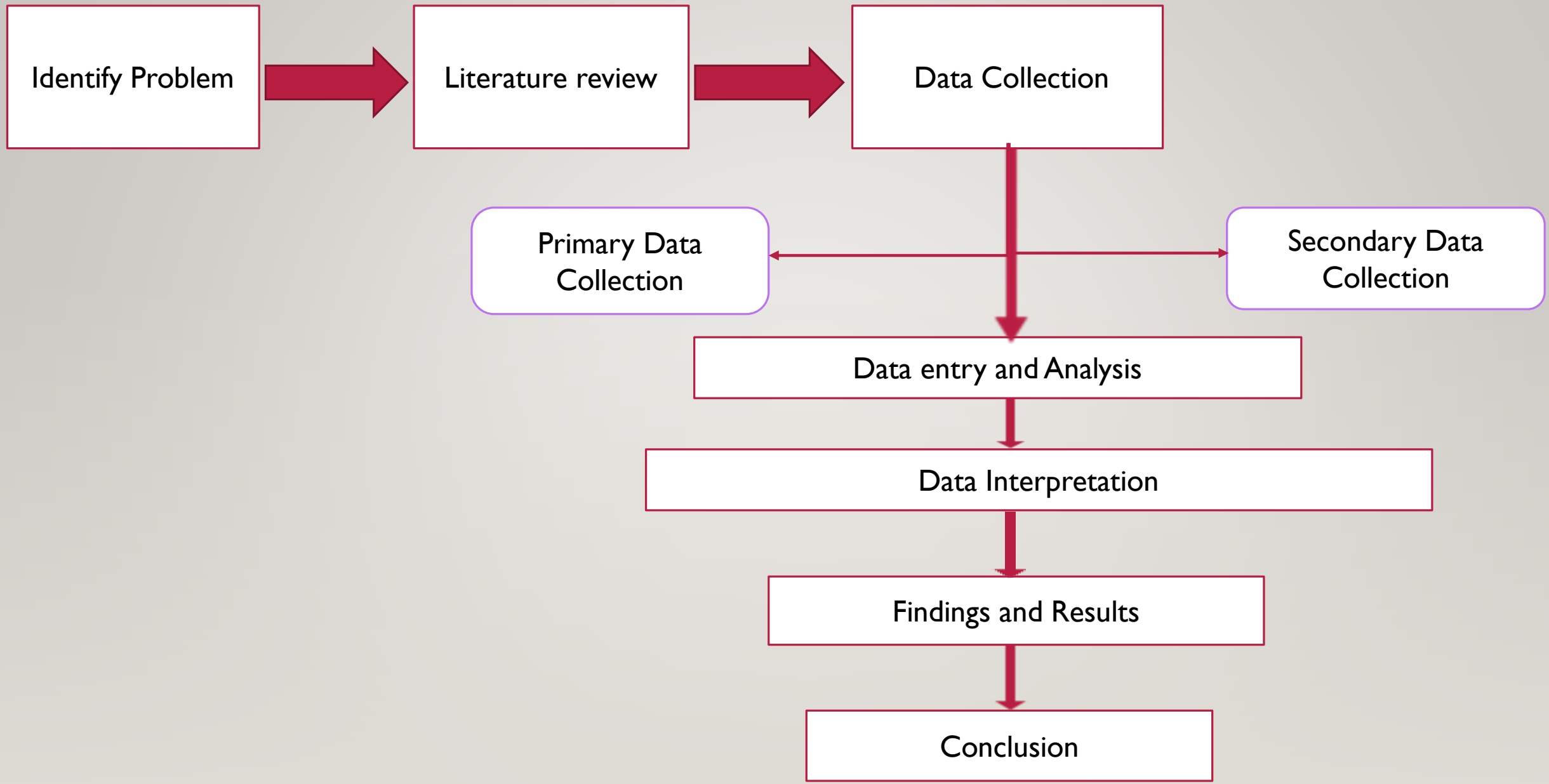
At this point, the respondents are questioned to be answered what their concepts of electricity as a resource. The questions were about the value of electricity as a resource, why he/she thinks electricity is a valuable resource, the power reserves, raw materials for the production of electricity, demand for electricity, the electricity production process, cost of electricity, etc.

- **'Sense' of Environment and Policy:**

Aiming to evaluate people's 'sense' on the 'environment' regarding electricity consumption, the respondents were asked some questions about fossil fuel, environmental pollution by energy production, environmental degradation, climate change impacts on increasing power consumption, global warming, current electricity policy, prepaid meters, renewable sources of energy, etc.

- **Statistical Analysis:**

For statistical analysis, Descriptive statistics(frequencies, percentages)were computed. For data entry, the IBM SPSS Statistic tool is used and for analyzing data, the 'Chi-square method' has been used. All the results and findings are then interpreted and a remarkable conclusion has been given at the end.


Identify Problem
Literature review
Data Collection
Primary Data Collection
Secondary Data Collection
Data entry and Analysis
Data Interpretation
Findings and Results
Conclusion

RESULTS AND FINDINGS

DEMOGRAPHIC ANALYSIS

Variable	Frequency	Percent
Gender		
Male	185	61.5
Female	116	38.5
Age		
18 to 30	252	83.7
31 to 40	30	10.0
41 to 50	12	4.0
51 to 60	4	1.3
61 to 70	3	1.0
Occupation		
Student	121	40.2
Government Job	24	8.0
Businessman	38	12.6
Housewife	13	4.3
Private Job	59	19.6
Unemployed	43	14.3
Other	3	1.0
Education		
Postgraduate	92	30.6
Graduate	188	62.5
Below HSC	11	3.7
Below SSC	4	1.3
Illiterate / Primary Education	1	.3
Other	5	1.7
Family Member		
1 to 2	11	3.7
3 to 5	217	72.1
6 to 8	66	21.9
9 to 10	4	1.3
More than 10	3	1.0

ASSOCIATION BETWEEN USE OF EXPENSIVE ELECTRIC APPLIANCES AND DIVISIONS

The p values show that Washing Machine and Iron is associated with Divisions.

Expensive Electric Appliances	Divisions	P value
AC	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.180
Refrigerator	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.427
Iron	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.698
Water Heater	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.956
Washing Machine	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.000
Oven	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.736
Fan	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.238
Rice Cooker	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.000

ASSOCIATION BETWEEN USE OF EXPENSIVE ELECTRIC APPLIANCES AND DIVISIONS

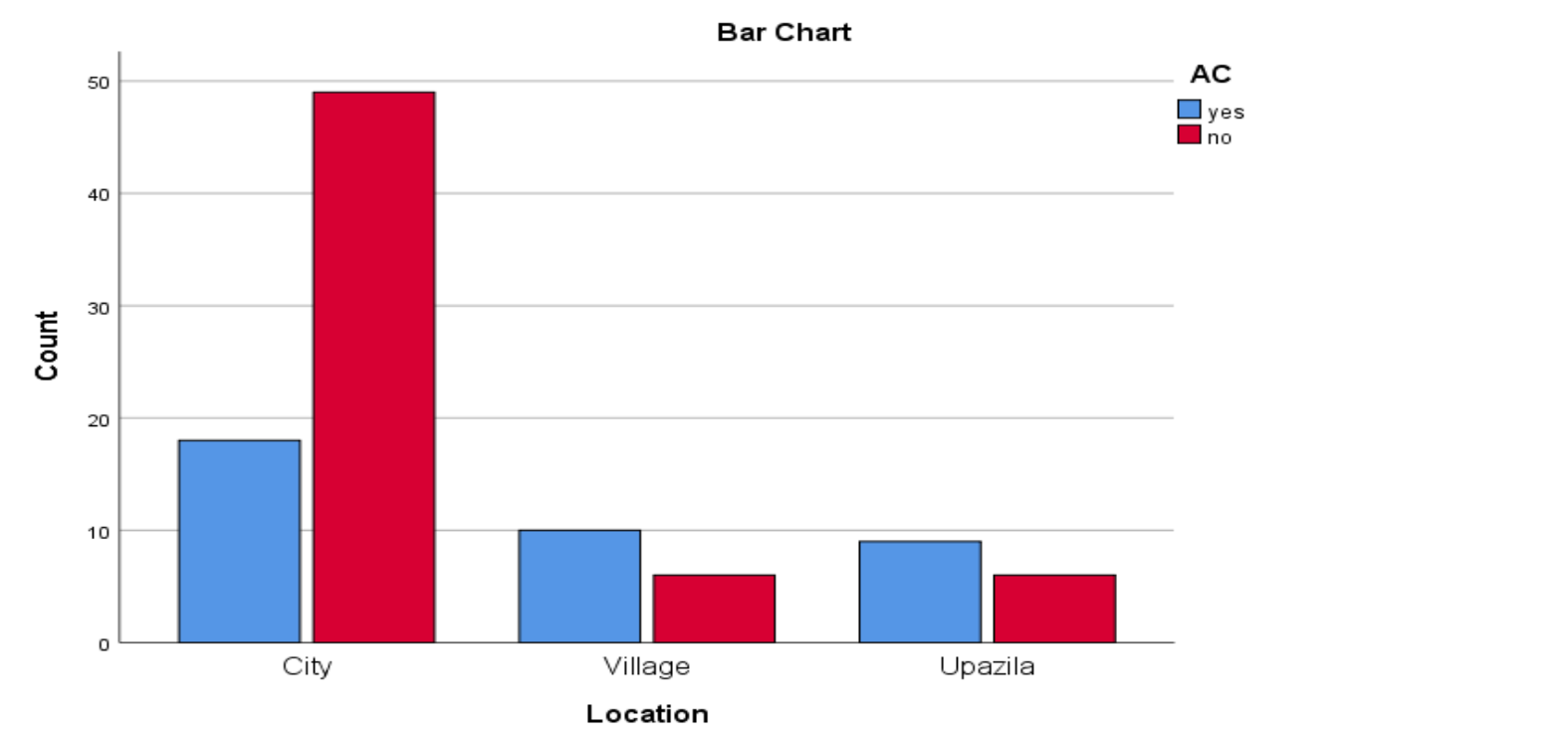


Fig: 1 (Association between location and usage of Air conditioner)

ASSOCIATION BETWEEN USE OF EXPENSIVE ELECTRIC APPLIANCES AND DIVISIONS

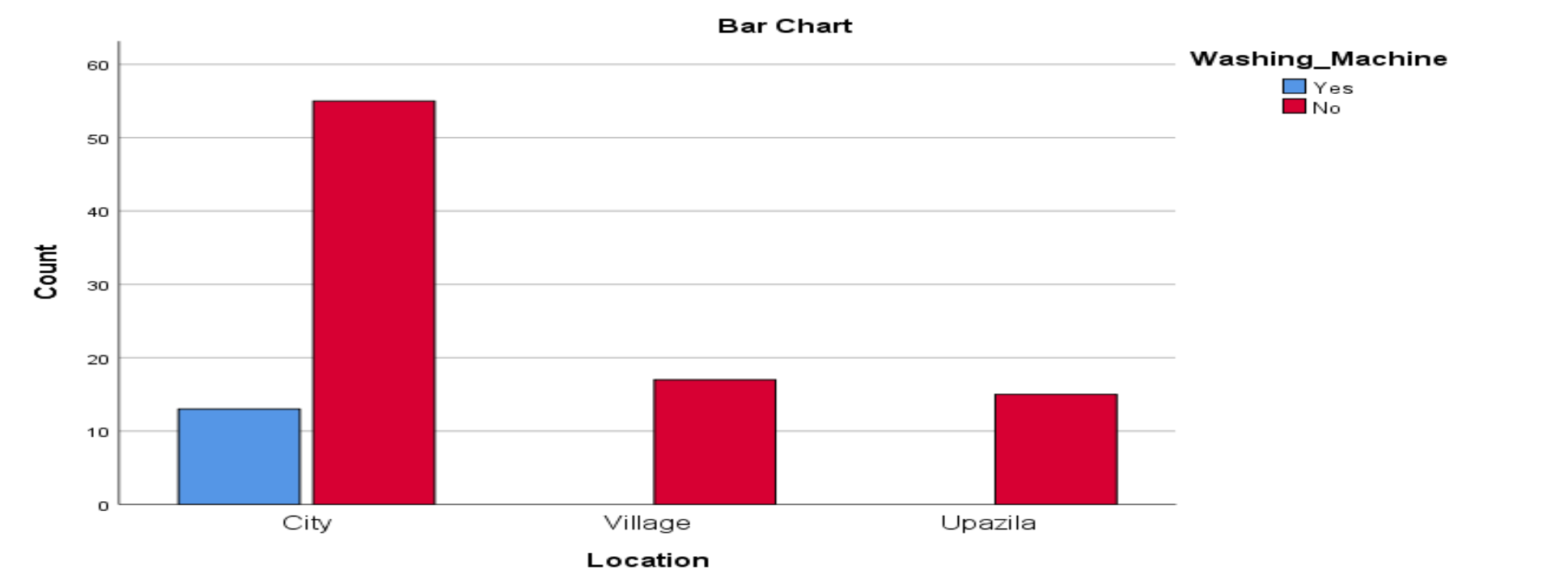


Fig:2 (Association of Washing Machine and Location)

CONCLUSION:

It is expected that the findings of the study will contribute to devising such an energy policy for the country that will promote a win-win nexus between electricity production, consumption, economic growth and environmental issues.

ASSOCIATION BETWEEN USE OF EFFICIENT ELECTRIC APPLIANCES AND DIVISIONS

Efficient Electric Appliances	Divisions	P Value
Energy Light	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.026
Refrigerator	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.000
Fan	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.000
Television	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.000
Rice Cooker	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.105
Washing Machine	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.353
Oven	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.044
Computer	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.001
AC	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.133
Iron	Chittagong, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, Barisal, Sylhet	.000

ASSOCIATION BETWEEN USE OF EFFICIENT ELECTRIC APPLIANCES AND DIVISIONS

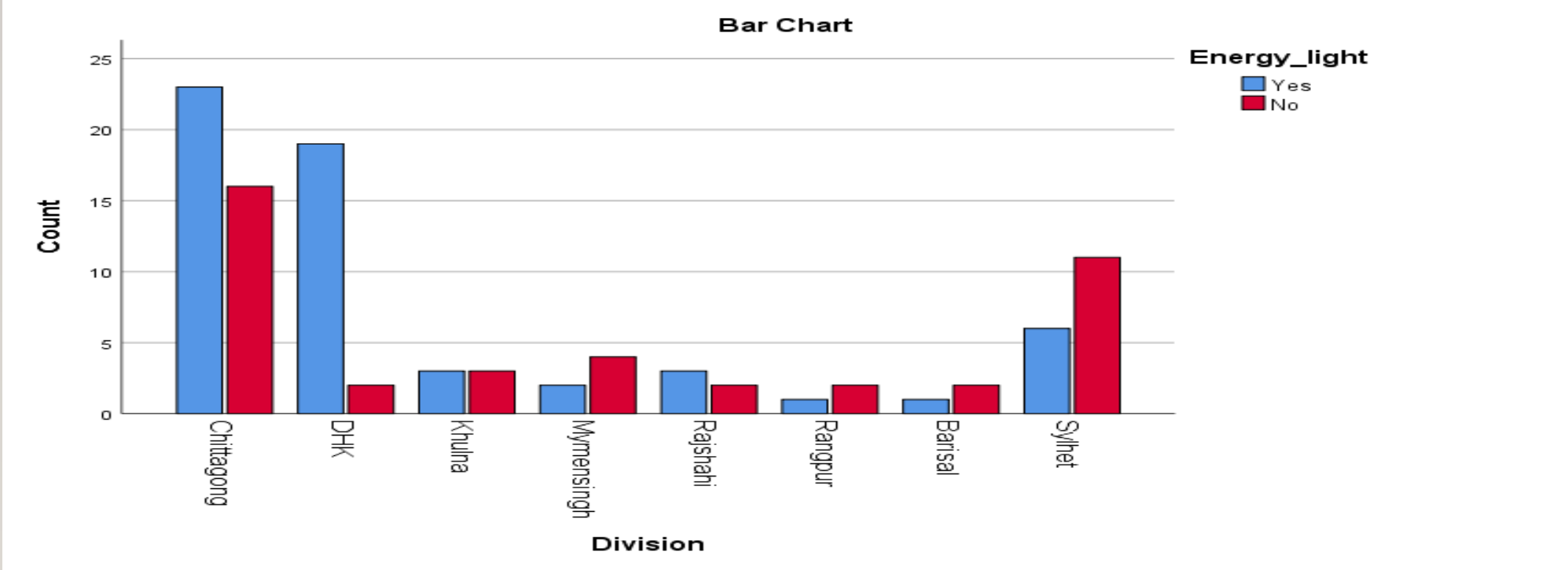


Fig: 3 (Energy light and Divisions)

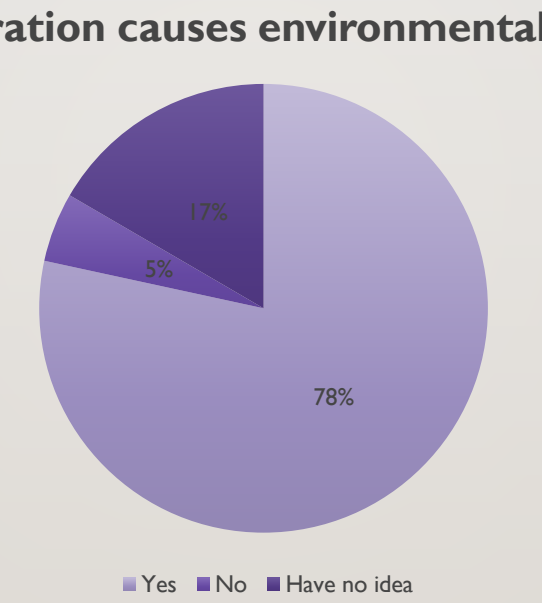
ASSOCIATION BETWEEN USE OF EFFICIENT ELECTRIC APPLIANCES AND LOCATION OF RESIDENCE?

AC is 6.6 times more associated with the location of the residence than the oven.

Efficient Electric Appliances	Location of Residence	P Value
Energy Light	City, Village, Upazilla	.604
Refrigerator	City, Village, Upazilla	.365
Fan	City, Village, Upazilla	.554
Television	City, Village, Upazilla	.130
Rice Cooker	City, Village, Upazilla	.644
Washing Machine	City, Village, Upazilla	.809
Oven	City, Village, Upazilla	.007
Computer	City, Village, Upazilla	688
AC	City, Village, Upazilla	.046

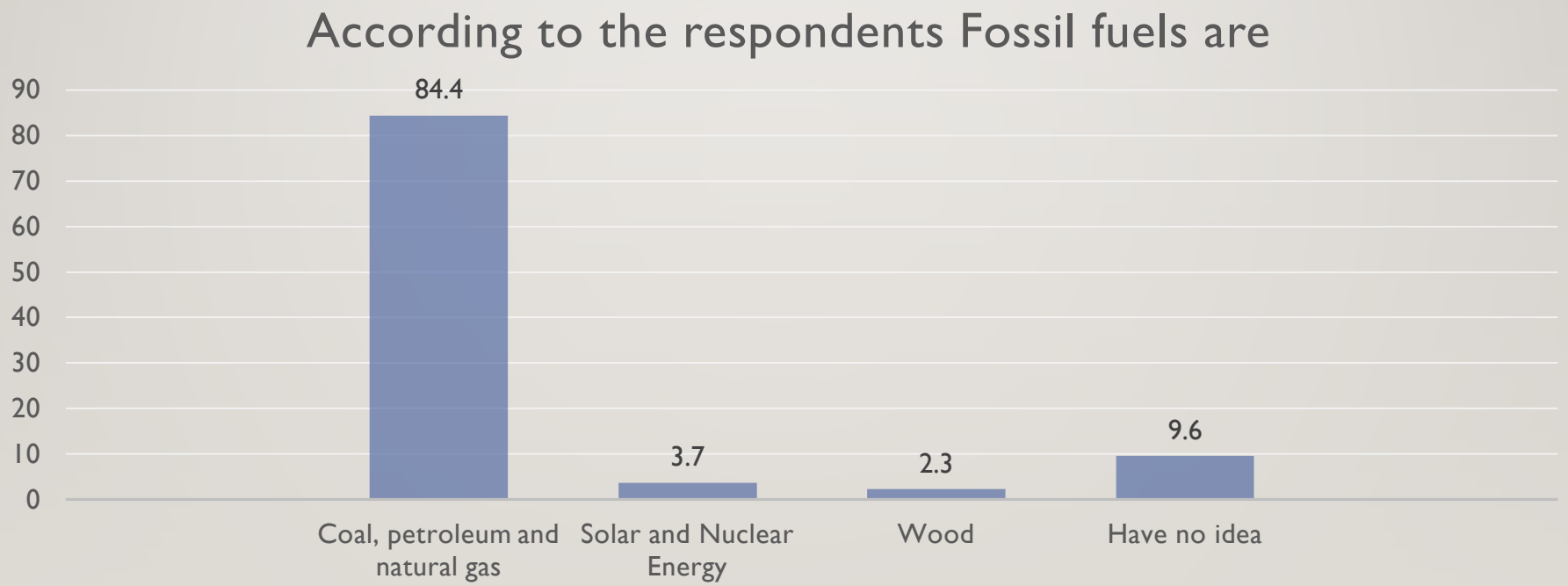
‘SENSE’ OF ENVIRONMENT OF THE RESPONDENTS

1. Is power generation causes environmental degradation?



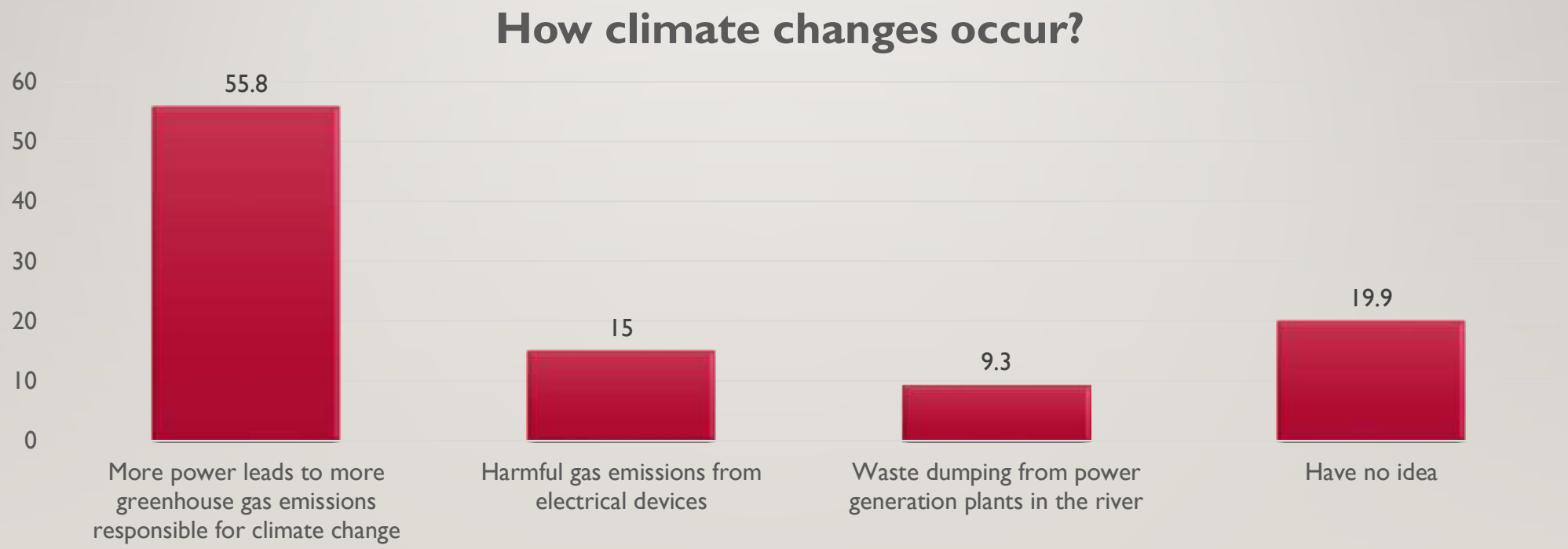
‘SENSE’ OF ENVIRONMENT OF THE RESPONDENTS

2. What are the Sources of Fossil Fuels?



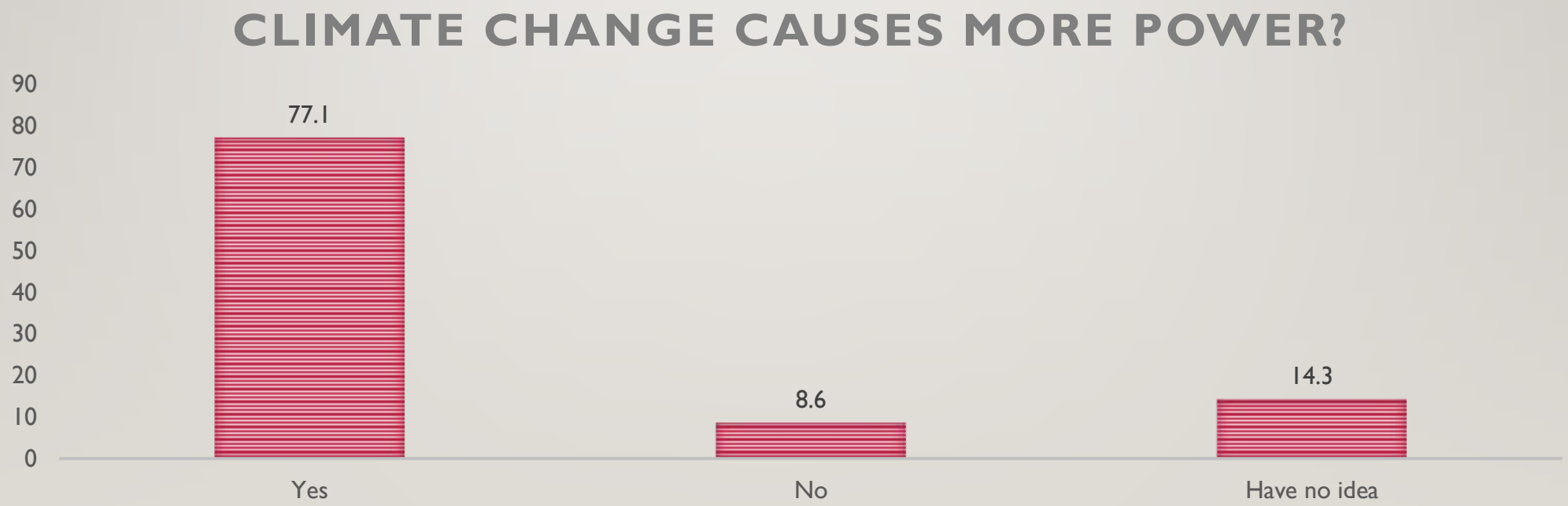
‘SENSE’ OF ENVIRONMENT OF THE RESPONDENTS

1. **How climate changes occur?**



‘SENSE’ OF ENVIRONMENT OF THE RESPONDENTS

1. **Climate change causes more power?**



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Electricity consumption and sense of 'environment' in everyday practice: a cross-sectional study in Bangladesh

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Abstract:

Bangladesh is a country with a promising economy despite its huge population and poverty. Electricity production and consumption play a significant role in ensuring economic development through secondary and tertiary economic activities for the countries like Bangladesh. However, awareness among the users and their overall attitude are the keys to efficient management of electricity production and consumption. On the other hand, environmental issues are a prime concern for Bangladesh when the country is considered one of the most vulnerable countries to climate change impacts. The present study examines users' attitudes and patterns of energy consumption in Bangladesh using a questionnaire survey. To assess the consumption pattern at the household level, secondary data have been collected from the annual report of the Bangladesh Power Development Board. Using the structured questionnaire survey, this research will analyze people's behavior and knowledge towards energy saving in day-to-day lives. In addition to focusing on the aforesaid issue, the study will take attempt to know how people think and how they act practically regarding electricity use. Findings of the questionnaire survey and the secondary data will be further analyzed using different statistical techniques like principal component analysis (PCA) to identify the most influencing factors of consumer behavior and how those are related to knowledge and environmental issues. It is expected that the findings of the study will contribute to devising such an energy policy for the country that will promote a win-win nexus between electricity production, consumption, economic growth and environmental management issues.

Keywords: Electricity, Development, Behavior, Environment, Bangladesh

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